

RAHUL K R

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Experience

Exotel Techcom Pvt Ltd.

July 2024 – Present

Member of Technical Staff 1 (Senior Engineer equivalent)

Bengaluru, Karnataka

- Led frontend development of the **Omnichannel Enterprise Contact Center** — a real-time voice and chat SaaS platform with integrated AI capabilities — now used by enterprise clients including QIIC, Bluedart, and Tafaseel.
- Drove UI performance optimization initiatives, implementing **code-splitting, caching, lazy-loading, and memoization**, resulting in **35% bundle size reduction** and measurable improvements in **Core Web Vitals**.
- Led **sprint planning, Architecture Decision Records (ADRs) creation** and monthly release delivery of 80% core product modules for a high-concurrency SaaS platform, maintaining under 10% delivery slippage.
- Conducted **root cause analysis (RCA)** for 100+ customer issues and delivered actionable **plans of action (POA)** and preventive measures, ensuring minimal service disruption, high customer satisfaction across enterprise accounts.

Exotel Techcom Pvt Ltd.

July 2022 – June 2024

Member of Technical Staff-1

Bengaluru, Karnataka

- Integrated UI with 7+ Spring Boot microservices via API Gateway (**REST, JWT auth**) and internal gRPC, consuming Kafka-originated events over WebSockets for real-time state synchronization across high-concurrency agent sessions.
- Monitored container health, pod statuses, and system metrics using **Kubernetes, ArgoCD, and Grafana/Prometheus**, building internal observability views to trace microservice failures and drive 99%+ service uptime.
- Architected a **micro-frontend** framework using **cross-domain messaging patterns**, enabling 3+ independent module deployment pipelines and secure runtime integration across legacy and modern UI applications.
- Engineered custom **keyboard navigation** loops and shortcuts across the core voice interface, improving **WCAG 2.1 accessibility** compliance and reducing agent task completion time.
- Participated in **code reviews, implemented proofs of concept (POCs)**, and mentored team members in adopting best practices, fixing UI-level vulnerabilities (**XSS**), and optimizing front-end performance.

Projects

PeerNest Mentoring Platform [Live Demo](#) | [GitHub](#)

April 2025 – Present

React.ts, Spring Boot, SockJS, STOMP, JWT, WebSocket, JPA, PostgreSQL, WebRTC, CursorAI

- Designed and implemented a Spring Boot backend using clean **MVC architecture** (Controllers, Services, Repositories) over PostgreSQL/JPA, with **JWT-based authentication** and role-based access control (RBAC).
- Developed **Mentor and Mentee dashboards** featuring **mentor discovery, session booking, real-time availability, session status tracking, and profile management**, delivering a seamless end-user experience.
- Integrated real-time communication capabilities using **WebRTC** for video conferencing and **WebSockets** for signaling and chat messaging, enabling dual video streams, screen sharing, and live session updates.

Emotion-Based Game for Cognitive Development of Children — Academic Project

2021 - 2022

Python, TensorFlow, Keras, CNN, OpenCV, Flask, REST API, NumPy, Pandas, Unity

- Built and trained a **Convolutional Neural Network** for facial emotion recognition using TensorFlow/Keras and the CK+48 dataset, achieving **90.6% accuracy** across 5 emotion classes.
- Implemented a **Flask REST API** to serve the trained model, handling image preprocessing (OpenCV, Haar cascade detection) and returning real-time inference results to a Unity game client.
- Engineered the client-server communication layer — image capture, encoding, and multipart data transfer — connecting the ML inference service to real-time gameplay logic.

Technical Skills

Languages: JavaScript (ES6+), TypeScript, Python, Java, HTML5, CSS3

Frontend Frameworks & Libraries: React.js, Next.js (SSR/SSG), Redux.js, React Router, Tailwind CSS, Material-UI, Web Workers, WebSockets, GWT, Responsive Design, WCAG 2.1/Accessibility, Core Web Vitals

Backend, Data & Cloud: Node.js, Kubernetes, Spring Boot, REST APIs, PostgreSQL, AWS, CI/CD, Linux

Developer Tools, Testing & Practices: Git, Vite, Webpack, NPM, Postman, Jest, React Testing Library, Playwright, Chrome DevTools, Figma, Prometheus, Grafana (RUM), ArgoCD, Swagger/OpenAPI, Jira, Cursor, Copilot, MCP

Awards

- Received **Certificate of Appreciation** from Exotel's **SVP of Engineering** for technical contributions and impact.
- Recognized in the company-wide **quarterly achievement list** for consistent high performance.

Education

National Institute of Technology, Calicut

April 2018 – June 2022

Bachelor of Technology (B.Tech), ECE – 7.63/10

Calicut, Kerala